

Hankel-Norm and Information Limits on Delay Detection by Low-Order State-Space Models, with Application to Graph Random Walks

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Abstract—An order- S linear time-invariant (LTI) state-space model must decide whether the current sample repeats the one k steps earlier, as random-walk graph learners do to detect cycles. For Gaussian inputs, exact detection requires $S \geq k + 1$ and a singular state matrix, so LTI invertible dynamics such as S4D (or Mamba with a fixed step) cannot achieve it. Since the lag- k delay has k unit Hankel singular values, for $S < k$ no linear probe beats the zero probe in Hankel norm, any probe explains at most S/k of the delayed sample's variance, and every readout has detection divergence at most $\frac{1}{2} \ln \frac{k}{k-S}$. Optimal fits track the S/k law, and a trained shift register switches on at $S = k + 1$. Trained diagonal SSMs need $4.0\times$ the minimal state to reach AUROC 0.95; frozen optimal poles remove most of this overhead for $k \leq 12$. On circular skip-link graphs the remaining gap lies downstream of the dynamics.

Index Terms—State-space models, Hankel operators, model order reduction, detection, random walks on graphs.

I. INTRODUCTION

DECIDING whether a signal repeats itself after a lag k is a basic detection task: it is echo detection, and it is the primitive that sequence models must implement to recall earlier inputs. Structured state-space models (SSMs) [1]–[3] implement it with a recursive filter of order S , and much of their design is about approximating delays with few states [4], [5]. That the pure delay z^{-k} has McMillan degree k is classical realisation theory [6]–[8]. What is missing is a *rate*: how much of the delayed sample can an order- S state carry when $S < k$, and what does that imply for detection by an arbitrary nonlinear readout? Related limits for recurrent models concern memory decay and optimisation [9], copying and associative recall [10], [11], and state tracking [12].

Our application is graph learning. Message-passing GNNs are bounded by the 1-WL test [13], [14] and cannot separate the regular circular skip-link (CSL) graphs [15], [16]. Walk-based learners read random walks as sequences [17]–[19], and random node features [20], [21] turn walk revisits, and hence closed walks, into token repetitions. SSMs are now used as the sequence backbone of such models [18], [22], [23].

Our contributions are: (i) *exact limits*: for Gaussian inputs, exact lag- k detection needs $S \geq k + 1$ and a singular state matrix, and is impossible for invertible dynamics at any S ;

(ii) *rates*: for $S < k$, no linear probe beats the zero probe in Hankel norm, the explained variance is at most S/k , and for Gaussian inputs this bounds the divergence available to every readout; (iii) *graph consequence*: exact non-backtracking (NB) return horizons $W^*(r)$ at which window features separate each CSL class; and (iv) *validation*: optimal fits, shift-register and frozen-pole controls, and seeded experiments that separate the information limit, the optimisation of the dynamics and the readout.

II. PROBLEM SETTING

A sequence model with real state $s_t \in \mathbb{R}^S$ reads tokens $x_t \in \mathbb{R}^d$:

$$s_t = A_t s_{t-1} + B_t x_t, \quad \hat{y}_t = g(s_t, x_t), \quad (1)$$

with an arbitrary readout g . It is *LTI* if $A_t \equiv A$ and $B_t \equiv B$. The *lag- k repeat* label is $y_t = 1[x_t = x_{t-k}]$. In the graph setting, $v_0, v_1, \dots$ is an NB walk ($v_{t+1} \neq v_{t-1}$), each walk draws fresh features $f_v \sim \mathcal{N}(0, I_d)$ i.i.d. over nodes, and $x_t = f_{v_t}$. Then $y_t = 1[v_t = v_{t-k}]$ almost surely, i.e. $y_t = 1$ exactly when the last k steps form a closed NB walk. Their frequencies follow from powers of the Hashimoto (NB) operator [24], [25].

III. LIMITS

Proposition 1 (Exact detection). *Let x_t be i.i.d. $\mathcal{N}(0, I_d)$ with a stationary past, and test $H_1: x_t = x_{t-k}$ against $H_0: x_t$ fresh, from (s_t, x_t) of a stable LTI model. (a) If $S \leq k$, or (b) if A is invertible (any S), every readout g has Bayes error > 0 . (c) If $S \geq k + 1$, a singular A suffices: the shift register $s_t = (w^\top x_t, \dots, w^\top x_{t-S+1})$ with $g = 1[w^\top x_t = (s_t)_{k+1}]$ has zero error for every $w \neq 0$.*

Proof: Under both hypotheses (s_t, x_t) is zero-mean Gaussian, so the two laws are either equivalent (error > 0) or mutually singular. The support under H_1 lies in that under H_0 , where x_t is free given s_{t-1} . So singularity needs $a^\top s_t + b^\top x_t = 0$ a.s. under H_1 but not under H_0 . Since s_{t-1} has the same law under both, this means $a^\top A s_{t-1} = v^\top x_{t-k}$ a.s. with $v = -(B^\top a + b) \neq 0$, i.e. $a^\top A^m B = \delta_{m,k} v^\top$ for all $m \geq 1$: an exact FIR realisation of $v^\top z^{-k}$ by (A, AB, a) . (a) Its Hankel matrix has rank k , so $S \geq k$. At $S = k$ the realisation is minimal, hence nilpotent and controllable through AB ; but $[AB, \dots, A^k B] = A[B, \dots, A^{k-1} B]$ has rank below k for nilpotent A . (b) If A is invertible, $a^\top A^{k+1} B = 0$ for all $j \geq 0$ makes a orthogonal to $A^{k+1} \mathcal{R} = \mathcal{R}$, the reachable space, so $v = 0$. (c) Under H_0 , $w^\top x_t \neq w^\top x_{t-k}$ a.s. ■

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Code and data: <https://github.com/The-MathKing/ITE>.

At $S = k$ the error in (a) can nonetheless be made arbitrarily small (Theorem 1), so the threshold $S \geq k + 1$ concerns exactness, not a practical barrier.

Exactness over *all* token sequences is stronger: it needs $S \geq (k + 1)d$, because the subspaces $A^j B \mathbb{R}^d$, $j = 0, \dots, k$, must be independent.

Every S4D discretisation $A = \exp(\Delta\Lambda)$, and Mamba's with a fixed step, is invertible, so exact detection is never available to these LTI models. The relevant question is how *well* an order- S state can carry the lag- k sample. A linear probe $z_t = c^\top s_t + \beta^\top x_t$ estimates $w^\top x_{t-k}$ with $\|w\| = 1$. Its error $z_t - w^\top x_{t-k} = \sum_{m \geq 0} e_m x_{t-m}$ has taps $e_m = c^\top A^m B - \delta_{m,k} w^\top$ for $m \geq 1$ and transfer function $E(z) = \sum_{m \geq 1} e_m z^{-m}$, with $\|E(e^{j\omega})\|$ the Euclidean norm of the $1 \times d$ row. Let $\mathcal{H}_k(f)$ be the $k \times kd$ block-Hankel section with (i, j) block f_{i+j+1} , $0 \leq i, j < k$.

Theorem 1 (Hankel limits). *If $S < k$, every LTI probe satisfies*

$$\|\mathcal{H}_k(e)\|_2 \geq 1, \quad \text{hence} \quad \sup_\omega \|E(e^{j\omega})\| \geq 1, \quad (2)$$

$$\|\mathcal{H}_k(e)\|_F^2 = \sum_{m=1}^{2k-1} \min(m, 2k-m) \|e_m\|^2 \geq k - S. \quad (3)$$

Hence, for i.i.d. tokens with $\mathbb{E}[xx^\top] = I_d$ and a stationary past (or $t \geq 2k-1$), $\mathbb{E}[(z_t - w^\top x_{t-k})^2] \geq 1 - S/k$: the probe explains at most a fraction S/k of the variance. For $S = k$ the infimum of the error is 0, but it is not attained.

Proof: Since $h_{i+j+1} = (c^\top A^i)(A^{j+1}B)$, we have $\text{rank } \mathcal{H}_k(h) \leq S$. The target $\delta_{m,k} w^\top$ gives a block anti-diagonal matrix with k unit singular values. The Eckart–Young–Mirsky theorem [26], [27] in the spectral and Frobenius norms gives $\|\mathcal{H}_k(e)\|_2 \geq \sigma_{S+1} = 1$ and $\|\mathcal{H}_k(e)\|_F^2 \geq \sum_{i>S} \sigma_i^2 = k - S$. A finite section is bounded by the Hankel operator Γ_E , and $\|\Gamma_E\| \leq \|E\|_\infty$ because Γ_E is a compression of the Laurent operator of E . This gives (2); Nehari's theorem [28] sharpens it to $\|\Gamma_E\| = \text{dist}(E, H^\infty)$. The tap e_m occurs $\min(m, 2k-m) \leq k$ times, which gives (3) and the variance bound. For $S = k$, take $A = \gamma P$ with P the k -cycle permutation, $B = u_1 w^\top$, $c = \gamma^{-k} u_1$ and $\beta = -\gamma^{-k} w$. Then $e_m = \gamma^{m-k} w^\top$ for $m \in \{2k, 3k, \dots\}$ and 0 otherwise, so the error is $\gamma^{2k}/(1 - \gamma^{2k}) \rightarrow 0$ as $\gamma \rightarrow 0$. Zero error would be an exact FIR realisation, which the proof of Proposition 1(a) excludes at $S = k$. ■

Bound (2) says that for $S < k$ no LTI probe beats outputting zero in Hankel norm: all Hankel singular values of z^{-k} equal one, so its optimal Hankel-norm approximant of degree $< k$ is zero [7], [8]. Bound (3) is the rate. Under Gaussian inputs it extends to every readout.

Corollary 1 (Any readout). *Let x_t be i.i.d. $\mathcal{N}(0, I_d)$ and $S < k$. For every measurable g , the minimum mean-squared error of x_{t-k} given (s_t, x_t) is at least $d - S/k$. For H_1 : $x_t = x_{t-k}$ against H_0 : x_t fresh, the observations satisfy $\text{KL}(P_1 \| P_0) \leq \frac{1}{2} \ln \frac{k}{k-S}$, so no detector exceeds balanced accuracy $\frac{1}{2} + \frac{1}{2} \left(\frac{1}{4} \ln \frac{k}{k-S} \right)^{1/2}$.*

Proof: For jointly Gaussian variables the conditional mean is linear. Stacking d probes gives a target with kd unit Hankel singular values, and (3) yields total error $\geq d - S/k$.

Next, (s_t, x_t) is a function of (s_{t-1}, x_t) . These are independent under H_0 and have the same marginals under H_1 , so $\text{KL} = I(s_{t-1}; x_{t-k}) = -\frac{1}{2} \sum_i \ln(1 - \lambda_i)$, where λ_i are the squared canonical correlations. Relative to s_{t-1} , x_{t-k} is the lag- $(k-1)$ tap, with the $m = 0$ tap included. The $k \times k$ section with blocks $c^\top A^{i+j} B$, $0 \leq i, j < k$, again has a target with k unit singular values, and tap m occurs $\min(m+1, 2k-1-m) \leq k$ times. Hence $\sum_i \lambda_i \leq S/k$. By convexity the maximum puts all mass on one λ_i . Data processing and Pinsker's inequality complete the proof. ■

Remark 1. The bounds limit *precision*, not information. For $S < k$ the state can still correlate with the lagged sample, so statistics pooled over many tokens remain informative. The near-optimal realisation at $S = k$ needs gains γ^{-k} and is exponentially ill-conditioned (cf. [9]). Input-dependent (A_t, B_t) break the factorisation used above, so we test them empirically. On walks, revisits make tokens dependent. Eq. (2) is distribution-free; the variance and divergence rates use the i.i.d. surrogate, which we check in Section IV.

NB return horizons. Let $p_G(\ell) = \Pr[v_\ell = v_0]$ for the stationary NB walk (uniform on directed edges). We compute it exactly from the Hashimoto operator [24]. For a family $\{G_r\}$, let $W^*(r)$ be the smallest W such that $\max_{\ell \leq W} |p_{G_r}(\ell) - p_{G_{r'}}(\ell)| > \tau$ for all $r' \neq r$. It is the shortest window of revisit frequencies that separates class r . For the ten CSL classes ($n = 41$, skips $r \in \{2, 3, 4, 5, 6, 9, 11, 12, 13, 16\}$), $W^*(r)$ ranges from 3 to 8 and is unchanged for $\tau \in [10^{-4}, 3 \cdot 10^{-3}]$ (Table I). W^* is a horizon for *exact* access. An order- S reader with $S < W^*$ still explains up to S/ℓ of the lag- ℓ variance. We therefore expect pooled estimators to succeed below W^* , at a sample cost that grows as S/W^* shrinks, since each token carries divergence at most about $S/(2\ell)$.

IV. EXPERIMENTS

Models. Our learned SSM is one diagonal complex layer with S4D-Lin initialisation [2]. It has $\lfloor S/2 \rfloor$ complex modes plus one real mode if S is odd, so its real state dimension is exactly S . The dynamics are $A = \exp(\Delta\Lambda)$ and $B_t = \Delta B$, with Δ learned per mode (LTI) or set to $\Delta_t = \text{softplus}(W_\Delta x_t + v_\Delta)$ (input-dependent step, the Mamba-style gate). The readout sees $[\Re s_t, \Im s_t, x_t]$ and comparison features $(Ux_t - Vs_t)^{\odot 2}$, followed by a two-layer MLP. We use $d = 8$ features, AdamW with a cosine schedule, and 3 seeds; intervals are 95% t -intervals. All runs are on CPU, and a script regenerates every number and figure in this letter.

Delay realisation. We fit an S -state diagonal impulse response to $\delta_{m,k}$ ($m \geq 1$), keeping the best of four restarts (Fig. 1). All 16 fits with $S < k$ respect $\|e\|_2 \geq \sqrt{1 - S/k}$, with a median excess MSE of 0.11 over the bound. The error first drops to about 0.1 at a median of $S = 1.0k$, so the rate of Theorem 1 is nearly attained by direct fitting.

Token-level detection. We train on NB walks ($T = 64$) over 256 random 4-regular graphs ($n = 24$, unions of two Hamiltonian cycles) and test on 64 unseen graphs. Fig. 2 shows that the test AUROC for y_t rises smoothly with S . There is no sharp transition at $S = k$, where the AUROC is 0.69–0.72. The state needed for a given AUROC grows

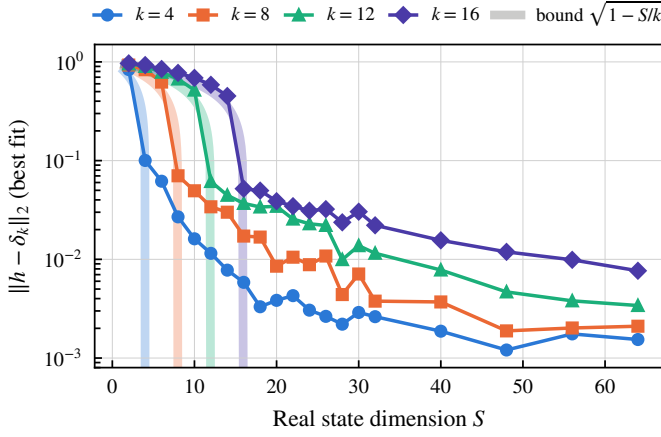


Fig. 1. Best gradient fit (four restarts, taps $m \leq 63$) of an S -state diagonal LTI impulse response to the lag- k delay. Shaded curves: the lower bound $\sqrt{1 - S/k}$ of Theorem 1 for $S < k$.

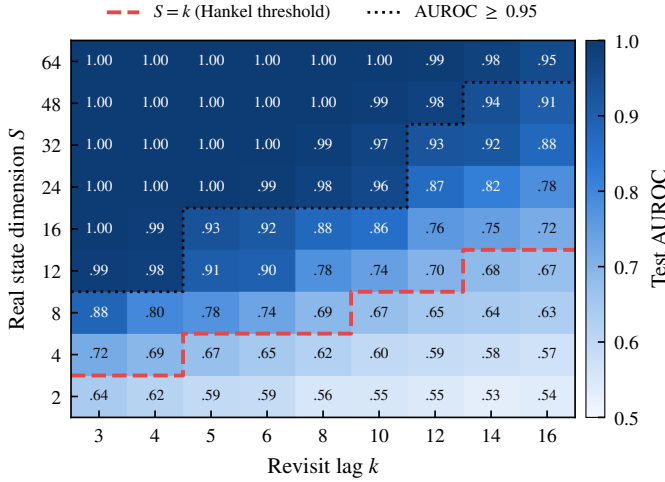


Fig. 2. Test AUROC of the LTI model for lag- k revisit detection on NB walks over random 4-regular graphs ($n = 24$, $T = 64$, $d = 8$; mean of 3 seeds). Red dashed line: $S = k$. Dotted line: first S with $\text{AUROC} \geq 0.95$.

approximately linearly in k . Its median ratio to k is 2.0, 2.7 and 4.0 at AUROC thresholds 0.8, 0.9 and 0.95; an AUROC of 0.99 is reached for only 7 of the 9 lags. Below the bound, all 22 LTI cells lie under the optimal AUROC of the Gaussian surrogate with explained variance S/k (Fig. 3). We checked the surrogate against walk structure with logistic regression on all pairwise coincidences among the S most recent tokens, which is what a shift register can compare. For $S \leq k - 3$ these predict lag- k revisits at AUROC 0.49–0.51, i.e. not at all. Within two lags of k they reach up to 0.53 ($S = k - 2$), 0.58 ($S = k - 1$) and 0.67 ($S = k$), because a closed walk that leaves and returns along the same path makes stored tokens coincide. The surrogate is therefore fair for $S \leq k - 3$. The input-dependent step changes the AUROC by -0.001 ± 0.002 (mean over cells, $|\Delta| \leq 0.03$; Wilcoxon $p = 0.005$, paired t -test $p = 0.14$), a negligible difference. Its learned Δ_t varies by only 1% across tokens, which is expected because random features give the gate nothing to select on.

Controls. Three controls separate the information limit from

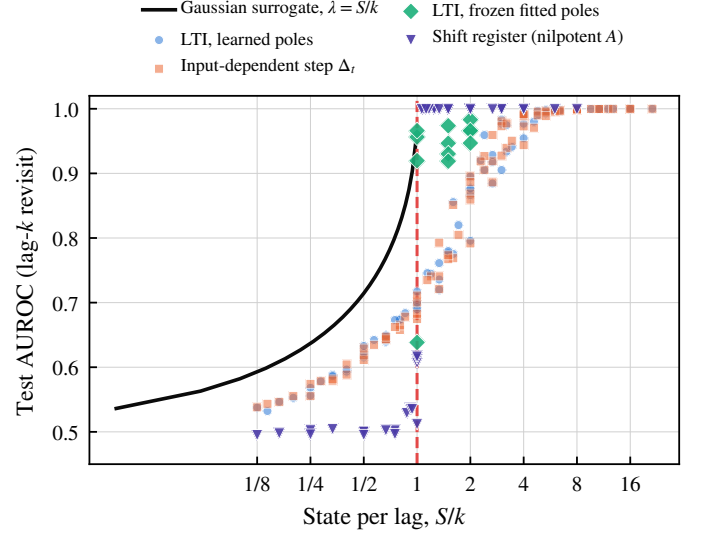


Fig. 3. Test AUROC against S/k (all k , seed means). Solid curve: optimal AUROC of the i.i.d. Gaussian surrogate with explained variance S/k (Corollary 1). Red dashed line: $S = k$.

training effects (Fig. 3).

- **Shift register.** A trained shift register (nilpotent A , learned w , $k \in \{4, 8, 12, 16\}$) has AUROC 0.51–0.62 at $S = k$ and 1.00 at $S = k + 1$. This is the switch predicted by Proposition 1(c). The excess over 0.5 at $S = k$ matches the pairwise coincidences above.
- **Frozen poles.** We freeze the poles at the delay fit of Fig. 1 and train only B and the readout. The AUROC is 0.87 at $S = k$ and 0.97 at $S = 2k$ (mean over k), against 0.70 and 0.85 with learned poles. For $k \leq 12$ it is already 0.92–0.97 at $S = k$. At $k = 16$ it drops to 0.64 ± 0.26 , where the fitted poles are ill-conditioned (Remark 1), and it reaches 0.95 only at $S = 2k$, half the state that learned poles need.
- **Comparison features.** Removing them changes the $k = 8$ AUROC by at most 0.02 at any S .

For $k \leq 12$ the information limit is therefore nearly attainable at $S \approx k$, and most of the overhead of trained SSMS comes from learning the dynamics, not from the readout. At $k = 16$ the conditioning of the optimal realisation accounts for about half of the overhead.

CSL classification. CSL graphs are 4-regular and featureless, so 1-WL is provably at chance (10%). Each training example is one NB walk ($T = 128$) on a CSL graph, and the class is predicted from mean-pooled features. At test time, each graph is judged from 16 walks ($T = 256$). As an *exact-count reference* we use logistic regression on the exact revisit frequencies at lags $1, \dots, S$. It is not an upper bound. At $S = 2$ it is at chance, because NB walks cannot return at lags 1–2, while the LTI model reaches 24.0%. The reference reaches 98.3%. For all 10 classes, its onset of 80% class accuracy is the first tested $S \geq W^*(r)$ (Table I). The trained models reach $60.4 \pm 7.9\%$ (LTI) and $55.2 \pm 8.5\%$ (input-dependent step) at $S = 64$. The largest tested S gives the best accuracy for both, and the curves are still rising. Neither remedy we tried

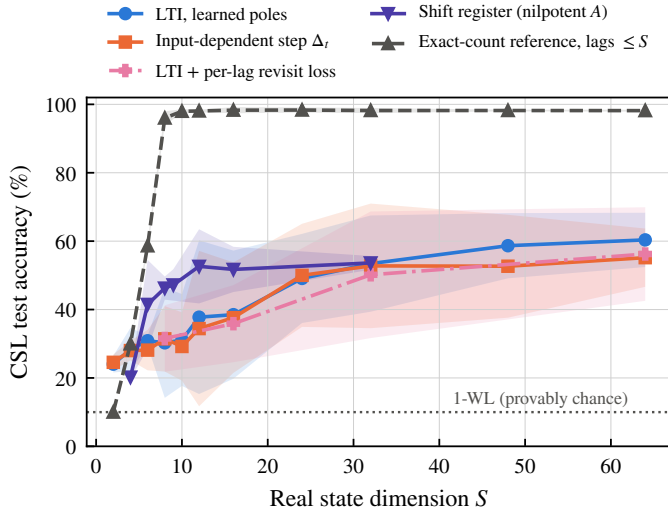


Fig. 4. CSL test accuracy against S (mean and 95% interval over 3 seeds; test graphs are the ten CSL graphs with fresh walks and features). The exact-count reference uses revisit frequencies at lags $\leq S$, which by Proposition 1(c) corresponds to an $(S+1)$ -state register.

TABLE I

CSL HORIZONS $W^*(r)$ AND PER-CLASS ONSETS: THE SMALLEST TESTED S WITH CLASS ACCURACY $\geq 80\%$ (MEAN OF 3 SEEDS; —: NOT REACHED FOR $S \leq 64$).

r	2	3	4	5	6	9	11	12	13	16
$W^*(r)$	3	4	5	6	7	8	7	8	5	7
Exact counts	4	4	6	6	8	8	8	8	6	8
LTI	2	6	—	—	—	—	—	—	32	—
Input-dep. Δ_t	2	8	—	—	—	—	—	—	64	—
Shift reg.	4	6	—	—	—	—	—	—	6	—

closes the gap:

- A per-lag revisit loss gives $56.2 \pm 13.7\%$ at $S = 64$.
- Evaluating at the training length ($T = 128$) instead of $T = 256$ lowers accuracy by 2.5 points.

The input-dependent step is -1.7 ± 1.1 points from LTI (paired over S and seeds). The dynamics are not the bottleneck either. A trained shift register has exact access to all lags $\leq S - 1$, with the same readout, pooling and labels. It reaches only 47.2% at $S = 9$, where the reference with lags ≤ 8 reaches 96.0%, and $53.6 \pm 2.0\%$ at $S = 32$. On CSL the gap therefore lies downstream of the dynamics, in the readout and mean pooling. Graph-level labels alone do not explain it, since the LTI model with a token-level revisit loss does no better. The shift register still tests Proposition 1(c) at the graph level: it first reaches 80% on classes $r = 2, 3$ and 13 at $S = 4, 6$ and 6 (Table I). This is the first tested $S \geq W^*(r) + 1$, the smallest register that holds lag $W^*(r)$. Class $r = 2$ ($W^* = 3$) is learned at $S = 2$ only by pooling: with one walk its accuracy is 53.7%, and with 16 walks it is 86.3%.

V. CONCLUSION

The pure delay has k unit Hankel singular values. For an order- S LTI state this gives three limits: for Gaussian inputs, exact detection needs $S \geq k+1$ and a singular state matrix; the

explained variance of the lagged sample is at most S/k ; and every readout has divergence at most $\frac{1}{2} \ln \frac{k}{k-S}$. Optimal fits and a shift register attain these limits. Trained diagonal SSMs respect them but need several times the minimal state. For $k \leq 12$, frozen optimal poles remove most of this overhead. On random-feature inputs, where the gate stays inactive, an input-dependent step does not change it. On CSL graphs, even exact lag access through a shift register leaves trained models far below exact counts, so there the bottleneck lies downstream of the dynamics, in the readout and pooling. Lower bounds for input-dependent dynamics and better-conditioned delay parameterisations are natural next steps.

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